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To Make or Not To Make? That is The Question

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INTRODUCTION

Adam, a woodworking enthusiast, lives with his family on a small farmette in rural Pennsylvania. He has been interested in woodworking since high school when his work was entered in the "Fresh Wood Student's Competition." Though he did not win the overall competition, his entry did receive the merit award for craftsmanship. Many years later, Adam now crafts wooden items to use in his own home and as gifts for his friends and extended family. Going forward, he hopes to supplement his income during retirement by making items to sell.

BACKGROUND

Over the last five years, Adam has invested in woodworking machines, tools, and supplies. These include a table saw ($2,495), a 16-inch planer ($1,950), and an 8-inch jointer ($750), the latter he bought secondhand on Craigslist. As these machines produce a significant amount of sawdust/fine-dust particles, Adam had to purchase a dust collection system and retrofitted this in his woodshop, which is a barn on his property. The cost of this system was approximately $900, excluding the time it took for him to install it. While there are many other tools and machines Adam would like to buy, he thinks the cost is prohibitive, unless he uses the items frequently. In addition to these "big ticket" items, Adam has acquired a myriad of smaller tools and supplies, such as chisels, handheld sanders, safety glasses, dust masks, sandpaper, screws, glue, and clamps. The wood Adam uses is mostly hardwoods that are native to Pennsylvania. He buys this locally for cash and his current timber "stack" costs in the region of $2,400.

IMMINENT OPPORTUNITIES



Cutting Boards for Realtor

A local realtor has approached Adam with a business proposition. Suzie heard about Adam through word of mouth when she admired a cutting board he made for a mutual friend. She would like him to make one-off/distinct cutting boards for her clients as a house-warming gift when they use her services to purchase a home. She estimates

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that she would need 50 cutting boards per year and will pay between $35 and $45 depending on the size of the board (small or medium). Before committing to a long-term contract, Suzie and Adam have agreed that she will purchase an initial batch of 10 boards that will be ready in a month.

Adam does his woodworking projects after work or on the weekends and does not have anyone to help him. He has all the materials on hand to make the batch of 10 boards, such as cut-off/scrap wood from other projects and glue, oil, and wax. He estimates that it will take three hours to make each board; the most labor-intensive element is sanding at roughly two hours. He is thinking of a number of alternatives for sanding:

1. Buy a drum sander, which would cost $1,250, and would cut his sanding time to one eighth.
2. Ask a semi-retired neighbor to do the sanding, using Adam's woodshop, supplies, and sanders, and pay $10 per hour—the same amount the neighbor earns in a current part-time job.
3. Take the boards to a local Amish woodshop that is a 45-minute drive (35 miles away). He would pay to use their drum sander—$50 per hour (in increments of one hour or part thereof)—that an employee would operate. This price includes all supplies.

The other hour to make the boards requires work on the table saw, planer, and jointer (both before and after gluing), clamping, and finishing the boards with food-grade oils and wax.



Headboard for a Friend

A good friend of Adam's asked him about making a bespoke headboard for her aunt. She has perused his timber "stack" and picked out the slab that she would like him to use, even though he has not yet committed to making it. She chose one of the largest and more unique pieces of live-edge walnut purchased as a "lot" of 15 slabs that cost $1,200.

Adam estimates that it would take 15 to 20 hours to make the headboard with approximately 75% of this spent on sanding it using the tools he currently owns. His sanding time, however, could be reduced considerably if he were to use one of the three options outlined previously. He knows that if he has to do all of the work himself on both projects, it is extremely unlikely he will complete them on time. Moreover, even if he could complete both projects, he does not have any idea how he would price the headboard.

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THE FUTURE

Longer term, Adam is thinking about converting his barn into a brick-and-mortar shop, which would open up his woodwork projects to the public. The barn is big enough to convert the front portion of the building into a reception area that could display some of his best work. This would allow the back portion of the barn to be devoted to woodworking.

Another potential opportunity came about when Adam's friend who lives in Philadelphia shared some images and a short video on social media. These featured Adam crafting cabinets and countertops for his own kitchen. After seeing the post, a real estate developer made inquiries and talked briefly with Adam about providing custom cabinets and/or butcher-block countertops within the development's luxury homes. While Adam likes the idea of opening his own brick-and-mortar shop, as it could allow him to be more creative, operating as a business-to-business (B2B) firm could potentially be more lucrative.

SUMMARY

Adam is in a quandary and needs professional help. He has reached out to your firm for assistance on how to cost his products to enable him to make informed decisions. He is also wondering if his woodworking endeavor could provide him with an income stream during retirement.

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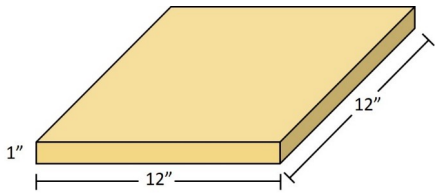
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Table 1. Additional Information from Adam

Supplies for cutting boards	Cost	Projected amount required/usage, etc.
Maple	$4 per board foot * see below	Approximately 60% of each board will be maple
Cherry	$6 per board foot	Approximately 20% of each board will be cherry
Walnut	$8 per board foot	Approximately 20% of each board will be walnut
Glue - waterproof food grade	$30 per gallon	Enough for 50 small/40 medium boards
Sandpaper - for current sanders	$20 dollars for 50 disks	Three different grits required - one disk of each grit per two boards (small or medium)
Conditioning oil - food grade	$37.50 for an 8- ounce jar	Enough for 75 small/60 medium boards
Wax - food grade	$37.50 for an 8- ounce jar	Enough for 100 small/80 medium boards
Sandpaper - for drum sander	$10 per drum	Three different drums required - each is a different grit - each drum is enough for 50 small or medium boards
Utilities	Gas heating and electricity	Rough estimate - $0.75 per board

Note: when paying for wood you pay by the board foot. A board foot is 144 cubic inches and is equivalent to a board or slab that is 1-inch thick x 12 inches wide x 12 inches long

One Board Foot



However, when buying wood, boards can be any size. For example, one could be six feet long x 8 inches wide and 1 inch thick, and another board might be three feet long x 11 inches wide and 2 inches thick. Assume that Adam only buys boards that are 1 inch thick.

Sizes of cutting boards

Small = 14 inches long x 8 inches wide and ¾ inch thick finished (starts out at 1 inch thick)
Medium = 17 inches long x 10 inches wide and ¾ inch thick finished (starts out at 1 inch thick)

Additional waste (other than thickness) is approximately 20% due to, for example, cutting with a table saw and using a planer.

Supplies for Headboard	Cost	Projected Amount Required/Usage, etc.
Wood: live edge walnut slab	One of a lot of 15 that cost $1,200	One slab – a larger and more unique piece
Conditioning oil: nonfood grade	$17.50 for an 8-ounce Jar	Enough for five headboards
Wax: nonfood grade	$17.50 for an 8-ounce jar	Enough for eight headboards
Sandpaper for current sanders	$20 dollars for 50 disks	Three different grits required – three disks of each grit
Sandpaper for drum sander	$10 per grit	Three different drums required - each is a different grit - each drum is enough for 10 headboards
Utilities	Gas heating and electricity	Rough estimate - $5 for headboard
Other costs		
Average cost per mile for car	All costs including gas	$0.46

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